Tab 1

AI-Enabled Safety Management Software

Proactive Risk Reduction System

Final Scope Confirmation Document

As per Meeting Discussion – 14 February 2026

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1. Executive Overview

This document represents the finalized workflow, structure, and implementation scope discussed on **14 February 2026**.

The objective of this system is to digitally transform safety audits through: - Structured zone-based monitoring - Inventory-level inspection tracking - AI-assisted risk identification - Centralized reporting & control

We request formal confirmation to proceed as per the below finalized scope.

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2. System Architecture Overview

2.1 Role Hierarchy Structure

```
┌──────────────────┐
│    ADMIN         │
│ (Super User)     │
└──────────────────┘
        │
        ▼
┌──────────────────┐
│ COORDINATOR      │
│ (Planning)       │
└──────────────────┘
        │
        ▼
┌──────────────────┐
│ FIELD OFFICER    │
│  (On-Site)       │
└──────────────────┘
```

This structure ensures controlled workflow and accountability.

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3. System Flow Diagram

```
Project Setup
      ↓
Upload Site Plan (Drone View)
      ↓
Upload Floor Plans
      ↓
Create Zones (Room-Level Mapping)
      ↓
Mark Inventory inside Zone
      ↓
Auto Generate ID + QR Code
      ↓
Assign Field Officer
      ↓
Field Inspection Upload
      ↓
AI Comparison & Risk Detection
      ↓
Coordinator Review
      ↓
Final Report Approval (Admin)
```

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4. Visual Module Representation

4.1 Site Plan (Visualization Layer)

Description: - Full structure overview - Surrounding area visibility - Used for high-level project visualization

Note: Site plan connects to floor plans only when multiple buildings exist.

4.2 Floor Plan (Operational Layer)

Description: - Basement / Ground / Upper Floors - Used for zone creation - Works as structured mapping blueprint

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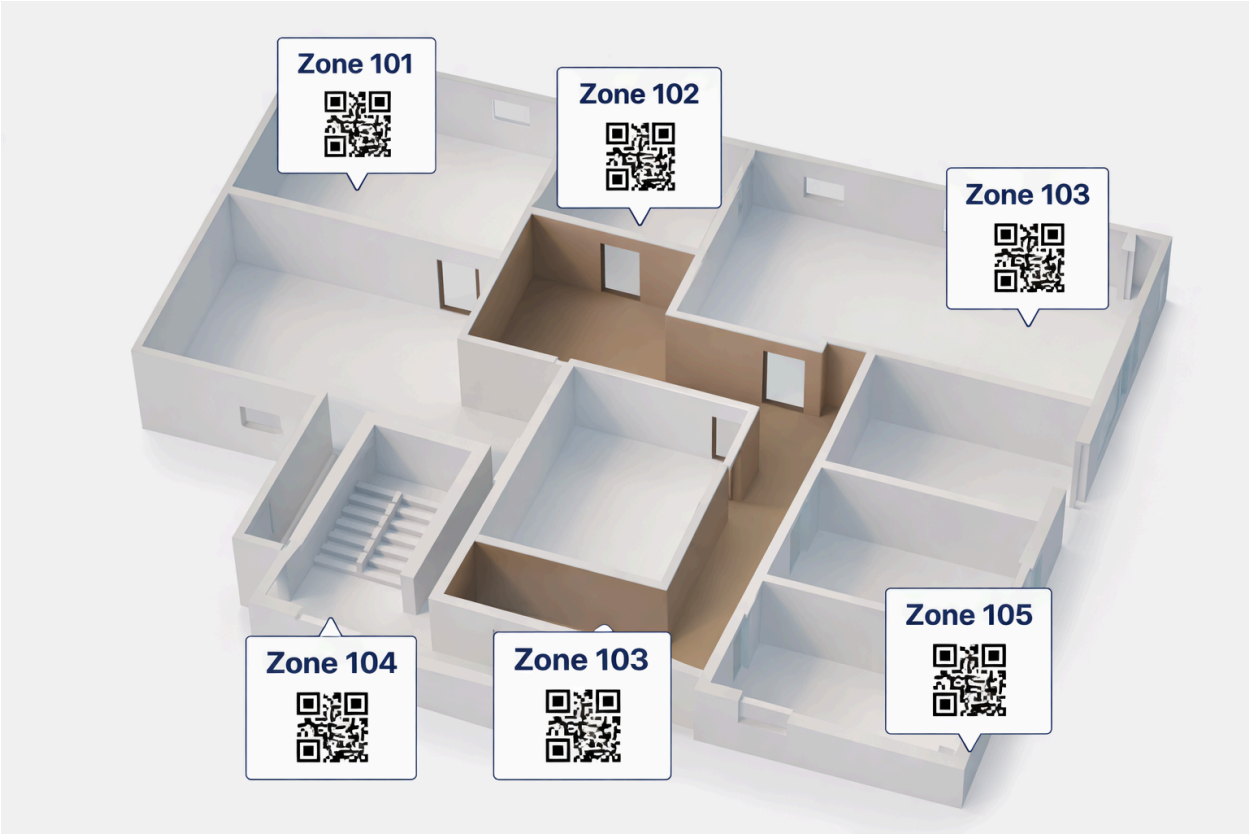
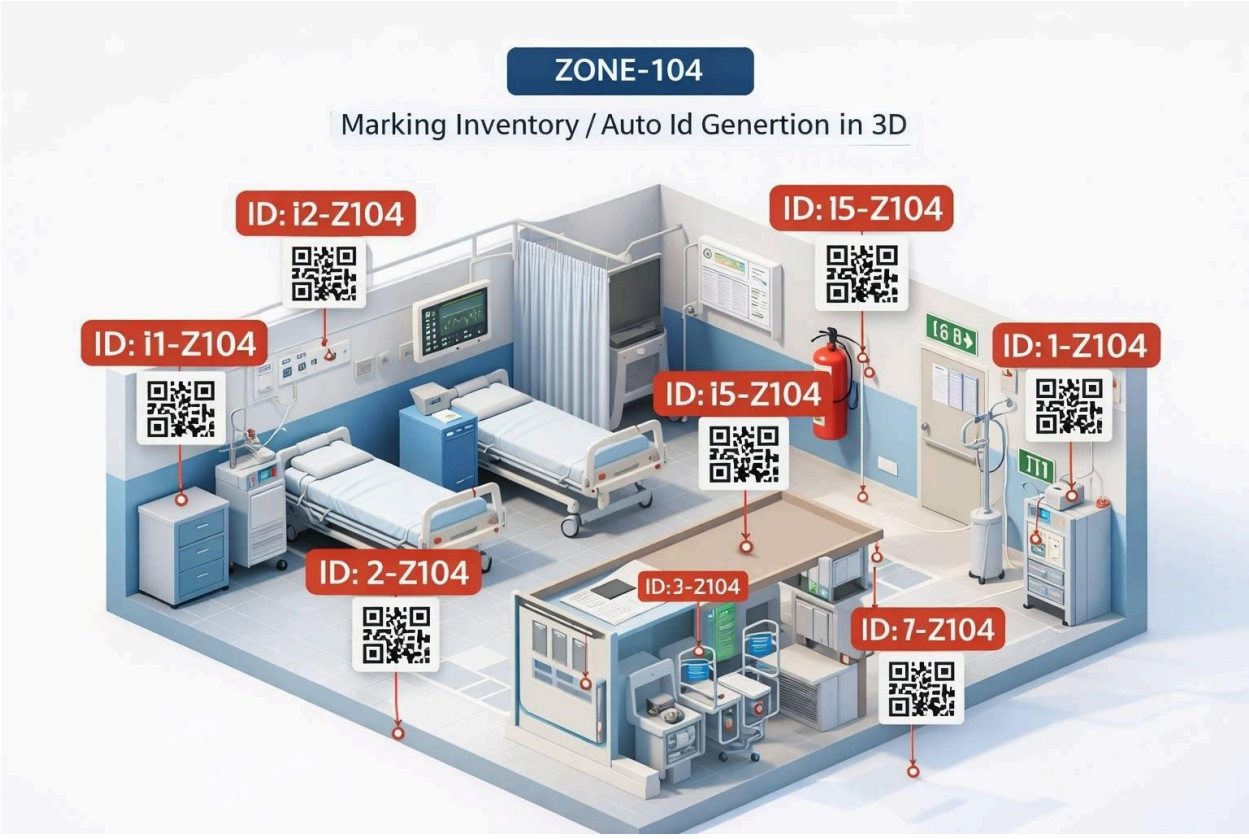
4.3 Zone Mapping



Zone Example: - Zone Name: 101 - Zone Type: Department / Sports / Accounts (Optional) - One Zone = One Primary Image or Video Reference

Zone = Defined Room / Area

4.4 Inventory Marking Inside Zone



For Each Inventory: - Auto ID Generated (I1-Z1) - Auto QR Code Generated - Upload Photo / Video / Description

Critical Assets Examples: - Fire Safety Systems - Emergency Exit Lights - Stair Safety - Safety Equipment

System Warning Triggered If: - Missing Inventory - Mismatch from baseline

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5. Role-Based Functional Responsibilities

5.1 Admin (System Owner)

- Full control access
- Coordinator management
- Project monitoring
- Final approval authority
- Report export (PDF / Excel)

5.2 Coordinator (Offsite Planning & Monitoring)

Responsibilities: - Organisation Setup (PD1) - Upload Site & Floor Plans - Create Zones - Mark Inventory - Generate QR Codes - Assign Field Officers - Review AI risk alerts - Submit Final Reports

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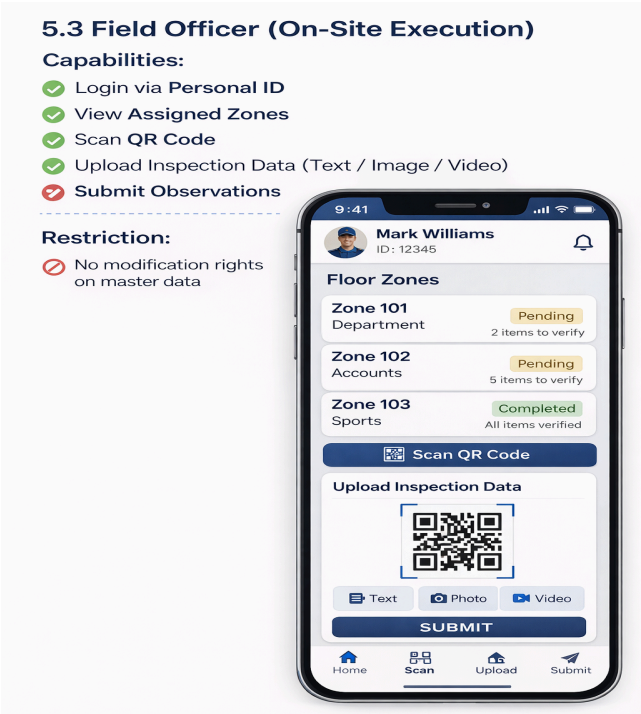
5.3 Field Officer (On-Site Execution)



Capabilities: - Login via Personal ID - View Assigned Zones - Scan QR Code - Upload Inspection Data (Text / Image / Video) - Submit Observations

Restriction: - No modification rights on master data

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6. AI Integration (Phase 1 Scope)

AI Functional Flow

```
Field Data Upload
        ↓
Compare With Baseline Inventory
        ↓
Detect Missing / Mismatch
        ↓
Generate Risk Flag
```

```
        ↓
Coordinator Review
```

AI will assist decision-making but will not auto-approve reports.

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7. Reporting Structure

Temporary Reports

- Daily
- Weekly
- Monthly
- Yearly

Final Report

- Coordinator Verified
- Admin Approved
- Exportable (PDF / Excel)

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8. Confirmed System Logic Rules

- 1 Coordinator → Multiple Organisations
- 1 Coordinator → Multiple Field Officers
- 1 Field Officer → Multiple Zones
- Zone = Room-Level Mapping
- One Zone = One Primary Media Reference
- Site Plan primarily for visualization
- QR-based inventory tracking mandatory
- Missing inventory triggers system alert

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9. Items for Final Client Confirmation

Kindly confirm the following:

1. AI validation limited to text & image comparison in Phase 1?
2. 3D visualization required in Phase 1 or Phase 2?
3. Mobile app required immediately or responsive web sufficient for MVP?
4. Admin approval mandatory for final submission?

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10. Development Phasing Plan

Phase 1 – Core System (Roles, Zones, Inventory, Assignment) Phase 2 – AI Risk Comparison Phase 3 – Advanced Analytics & Dashboard

Timeline will be finalized post approval.

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11. Conclusion

This document reflects the final discussion held on 14 February 2026.

Kindly review and confirm: - Alignment with your expectations - Any required modifications before development

Upon written confirmation, development execution will commence.

Prepared By: Project Team AI-Enabled Safety Management Software

Date: 16 Feb 2026

Tech working flow

AI-Enabled Safety Management Software

System Flow Diagram

```
Project Setup
      ↓
Upload Site Plan (Drone View)
      ↓
Upload Floor Plans
      ↓
Create Zones (Room-Level Mapping)
      ↓
Mark Inventory inside Zone
      ↓
Auto Generate ID + QR Code
      ↓
Assign Field Officer
      ↓
Field Inspection Upload
      ↓
AI Comparison & Risk Detection
      ↓
Coordinator Review
      ↓
Final Report Approval (Admin)
```

1️⃣ ROLE-BASED TASK BREAKDOWN

🔵 ADMIN ROLE (System Owner – Super Control)

✅ Core Responsibilities

1. User & Role Management

- Create / Edit / Delete Coordinators/ Filed Officers

- Assign Coordinators to Projects
- Manage Role Permissions (RBAC)
- Activate / Deactivate Role

2. Project Control

- Create Organisation
- Monitor Status (Pending / Ongoing / Completed)

3. Monitoring & Reports

- View all reports
- Zone-wise reports
- Export PDF / Excel
- Risk summary dashboard
- Final Reports

🛠️ Tech Implementation (Admin)

Feature	Tech
Role Based Access	Django RBAC + JWT
Authentication	JWT End-to-End
Dashboard	React + Chart.js
Report Export	Django + Python PDF (ReportLab) / Excel

Email Alerts	SMTP / SendGrid
Notification	Firebase FCM

🟢 COORDINATOR ROLE (Planning & Monitoring)

This is the most complex role.

✅ Responsibilities

1. Project Setup

- Upload Site Plan
- Upload Floor Plan
- Create Zones
- Create Inventory in Zones
- Auto Generate Inventory ID
- Auto Generate QR Code fill details
- Create personal ID to Filed-Officer
- Assign Field Officers(allot-zone)

2. Monitoring

- Compare AI detection old + new image

- Review reports
- Approve / Reject (items photo by filed-officers)
- Generate Temporary & Final Reports
- View inventory items
- Add more coordinators

🛠️ Tech Implementation (Coordinator)

Feature	Tech
Floor Plan Upload	Django + Cloudinary / AWS
Drawing Marking	Apryse OR Konva.js
3D View	Three.js / React Three Fiber
QR Generation	Python qrcode library
AI Comparison	OpenCV
Dashboard	React + Chart.js
File Upload	Django + DRF

🟠 FIELD OFFICER ROLE (On-site Execution)

✅ Responsibilities

- Login with Personal ID
- View Assigned Zones
- Scan QR Code
- Upload Data (Image)
- Submit Observations

⚠ Field Officer CANNOT edit inventory structure.

🛠️ Tech Implementation (Field Officer)

Feature	Tech
Mobile App	React Native
QR Scanner	React Native Camera
Upload Media	Django + Cloudinary
Notifications	Firebase FCM
Authentication	JWT

2️⃣ COMPLETE TECHNOLOGY STACK

Frontend (Web)

- React.js
- Chart.js
- Konva.js / Apryse
- React Three Fiber

Mobile

- React Native

Backend

- Python Django
- Django Rest Framework
- FastAPI (AI microservice)

Database

- PostgreSQL

Authentication

- JWT (End to End)

AI

- OpenCV (comparison logic)

Storage

- Cloudinary (MVP)
- AWS (Advanced phase)

Notification

- Firebase Cloud Messaging (FCM)

Email

- Django SMTP

3️⃣ PHASE-WISE EXECUTION PLAN (3 MONTH MVP)

You asked: Role based + Phase based.

We'll do:

- Phase 1 → Core System
- Phase 2 → Role Completion
- Phase 3 → AI integration + Reports+ Testing/Deploy

🟣 PHASE 1 (Month 1 – Foundation & Role Structure)

Goal:

Build authentication + role system + basic project setup.

What We Build:

✅ Backend

- Django project setup
- PostgreSQL schema
- JWT Authentication
- Role based access control
- User Management (Admin)
- Organisation Model
- Project Model
- Basic APIs (DRF)

✅ Frontend

- Login / Signup
- Admin Dashboard (basic)
- Create Coordinator
- Create Organisation
- Create Project

✅ Mobile

- Login
- Basic Dashboard

⏱ Duration: 4 weeks

🟣 PHASE 2 (Month 2 – Core Workflow Completion)

Goal:

Complete Admin +Coordinator & Field Officer Flow

Build:

Coordinator Side:

- Upload Site Plan
- Upload Floor Plan
- Create Zones using Konva
- Create inventory manual
- Auto ID generation
- QR code generation
- Assign zones to Field Officer

Field Officer:

- View assigned zones
- Scan QR
- Upload data (image/video)
- Submit report

Admin:

- View reports
- Monitor project status

⏱ Duration: 4 weeks

This completes FULL functional MVP (without AI).

🟣 PHASE 3 (Month 3 – AI + Reports + Testing + Deployment)

Now we separate properly as you requested.

🔴 PHASE 3A – AI Integration (2 Weeks)

Build:

- FastAPI microservice
- Integrate YOLO model/Open CV
- Image comparison logic
- Detect missing inventory
- Mismatch warning
- Send alert to Coordinator / filed Officer

Flow:
Field Officer Upload → AI Model → Compare DB → Flag Risk → Show Dashboard Alert

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🔴 PHASE 3B – Advanced Reports (1 Week)

- Temporary Report (Daily / Weekly / Monthly/Yearly)
- Final Report (Coordinator/admin verified)
- Risk Summary Dashboard
- Export Excel
- Email Report to Admin

🔴 PHASE 3C – Testing & Deployment (1 Week)

Testing:

- Unit testing
- API testing
- Mobile testing
- Security test (JWT validation)

Deployment:

- Backend → AWS EC2
- DB → AWS RDS
- Frontend → Vercel
- Mobile → Play Store Beta
- Media → Cloudinary

📅 TOTAL TIMELINE SUMMARY

Phase	Duration
Phase 1	4 weeks
Phase 2	4 weeks
Phase 3A	2 weeks
Phase 3B	1 week
Phase 3C	1 week

Total = 12 Weeks (3 Months MVP)

🔥 FINAL CLEAN FLOW (System Architecture)

Admin
↓
Coordinator
↓
Field Officer
↓
Upload Data
↓
AI Compare
↓
Risk Alert
↓
Report
↓
Admin Approval

working flow 3 month

Phase / Time	Focus Role	Backend & Database (Django + PostgreSQL)	Frontend (React.js + Konva)	Mobile App (React Native)	Key Technology	Final Output
Phase 1: Foundation (Month 1)	Admin	Auth System (JWT), RBAC Tables, Organization & Project Models, API Architecture.	Login/Signup UI, Admin Dashboard, User Management (Add/Edit Coordinators).	Mobile Login, Auth Integration, Splash Screen, Dashboard UI.	Django, PostgreSQL, JWT, React	Live Auth & User Control
Phase 2: Mapping (Month 2, W1-2)	Coordinator	Floor Plan Storage, Image Upload APIs (Cloudinary/S3), Zone-mapping DB logic.	Canvas Integration: Upload Plans, Draw/Create Zones, Mark Inventory.	Media (Camera/Gallery) Picker Integration, Upload Handlers.	Konva.js, Cloudinary, DRF	Digital Planning Tool

Phase 2: Execution (Month 2, W3-4)	Coordinator / Officer	QR Code Generator Engine, Task Assignment Logic, Field Officer Allotment DB.	QR Preview & Print Tool, Zone-to-Officer Assignment Interface.	QR Scanner, Assigned Zone List, Inspection Form (Photo/Video).	Python qrcode, RN Camera	Field Data Collection
Phase 3A: AI Core (Month 3, W1-2)	System (AI)	FastAPI Setup: YOLO Integration, Image Comparison Logic, Risk Scoring DB.	Comparison UI: Side-by-Side View (Old vs New), Highlight Detected Risks.	Real-time Push Notifications for flagged safety risks.	YOLO, OpenCV, FastAPI, FCM	Automated AI Risk Logic
Phase 3B: Reports (Month 3, W3)	Admin / Coord	PDF/Excel Export (ReportLab), Email Automation (SendGrid), Risk Statistics API.	Analytics Dashboard (Charts), Approval/Rejection Workflow UI.	User Submission History, Officer Performance Stats.	Chart.js, SendGrid, SMTP	Final Report Generation

Phase 3C: Launch (Month 3, W4)	DevOps	Performance Tuning, API Security Audit, AWS RDS & EC2 Deployment.	Production Build, Vercel/Netlify Deployment, UI Final Polish.	Firebase FCM Setup, Play Store Beta Testing, App Bundle Export.	AWS, Vercel, Firebase	Production Launch MVP